

Data Abstraction & 3-Level Schema Architecture

Q. What is Data Abstraction in a DBMS? Explain the three levels of data abstraction with a suitable example.

📌 Definition to Remember

Data Abstraction is the process of hiding the complex, low-level details of how data is stored and maintained, while exposing only the relevant, simplified information to end-users. It is implemented through the **ANSI/SPARC three-level architecture**: Physical, Logical, and View levels.

1. The Three Levels of Data Abstraction

VIEW LEVEL (External)	← End Users see their specific views
LOGICAL LEVEL (Conceptual)	← Designers see the full structure
PHYSICAL LEVEL (Internal)	← DBA sees raw storage details

1. Physical Level (Internal Level) — *Lowest Level*

- Describes **how** the data is actually stored on physical media (hard disks, SSDs).
- Handles: file organization methods (sequential, indexed, **hashed**), data compression, and encryption.
- Accessed by: **Database Administrator (DBA)** and system developers only.

2. Logical Level (Conceptual Level) — *Middle Level*

- Describes **what** data is stored and the **relationships** among that data.
- Defines the full database structure: tables, rows, columns, data types, **primary keys**, and **foreign keys**.
- Hides all physical storage complexities.
- Accessed by: **Database Designers** and **Application Programmers**.

3. View Level (External Level) — *Highest Level*

- Describes only the **specific part** of the database relevant to a particular user.
- Multiple users can have different **views** of the same database, hiding unauthorized or irrelevant data.
- Accessed by: **End-users** via application interfaces.

2. Example — University Database

Level	What is Seen
Physical Level	Records stored on Disk D in 256-byte blocks using a B+ tree index.
Logical Level	Table STUDENT (Roll_No, Name, Department, Fees) with data types and keys.
View Level (Professor)	Sees only Roll_No and Name — fees are hidden.
View Level (Accountant)	Sees only Roll_No , Name , and Fees — grades are hidden.

★ Must-Write Points (for 10 marks)

1. Data abstraction hides complexity and shows only relevant data to each user.
2. It follows the ANSI/SPARC three-level architecture: Physical, Logical, and View.
3. Physical level deals with actual storage — files, indexes, compression, encryption.
4. Logical level defines the full database structure: tables, columns, data types, keys.
5. View level provides user-specific views, hiding irrelevant or sensitive data.
6. Each level hides details from the level above — enabling **data independence**.
7. Example: A Professor sees student names; an Accountant sees fees — from the same database.

⚡ Quick Recall

Physical Level (how stored) → Logical Level (what stored + relationships) → View Level (user-specific view) → Data Independence